

RUIHAN YANG

✉ ruihan.yang@uci.edu  <https://buggyyang.github.io>  Ph.D. in Computer Science

Research Interests

Vision Language Models Diffusion Models Neural Data Compression

Education

Ph.D. in Computer Science

University of California, Irvine

2019 - 2025

B.S. in Computer Science

NYU Shanghai, New York University

2014 - 2018

Professional Experience

Member of Technical Staff

Nuance Labs Inc., Seattle

Jun 2026 - Now

- Multimodal foundation model and interaction model

Senior Researcher

LLM Frontier (ex-AI Lab), Tencent America, Bellevue

Apr 2025 - Jun 2026

- Main project: Multi-modal research on vision understanding (including vision language model and agent)
- Side project: text-to-speech generation

Internship Experience

Research Intern

Microsoft Azure AI, Microsoft, Redmond

Jun 2024 - Mar 2025

- Team: Cognitive Service Research & Voice AI
- Research on diffusion-based audio guided video translation (2024 summer) and talking avatar. Prototyping 2D diffusion-based human head simulation.

Research Intern

Microsoft Research, Microsoft, Redmond

Jun 2023 - Sep 2023

- Team: Audio and Acoustic Research
- Audio-Visual joint synthesis using multi-modal diffusion models

Interim Engineering Intern

Qualcomm AI Research, Qualcomm, San Diego

Jun 2021 - Sep 2021

- Team: Neural Data Compression Research
- Led the development of a pioneering project on end-to-end variable bitrate neural video compression

Research Assistant

Computer Science & Physics, NYU Shanghai

Jan 2018 - Jul 2019

- Research 1: Neural Music Modeling/Generation
- Published two papers as lead author and one as co-author at ISMIR and NIME conferences.
- Research 2: Applied Machine Learning & Scientific Computing
- Co-authored two papers published in Nature Communications and Journal of Physics: Condensed Matter

Publications

** denotes equal contribution*

Penguin-VL: Exploring the Efficiency Limits of VLM with LLM-based Vision Encoders

Boqiang Zhang*, Lei Ke*, **Ruihan Yang***, Qi Gao*, Tianyuan Qu*, Rossell Chen, Dong Yu, Leoweiliang
Technical Report, 2026

AstroCompress: A benchmark dataset for multi-purpose compression of astronomical data

Tuan Truong*, Rithwik Sudharsan*, Yibo Yang, Peter Ma, **Ruihan Yang**, Stephan Mandt, Joshua S. Bloom
ICLR, 2025

Fast Samplers for Inverse Problems in Iterative Refinement Models

Kushagra Pandey*, **Ruihan Yang*** and Stephan Mandt
NeurIPS, 2024

Precipitation Downscaling with Spatiotemporal Video Diffusion

Prakhar Srivastava, **Ruihan Yang**, Gavin Kerrigan, Gideon Dresdner, Jeremy McGibbon, Christopher Bretherton and Stephan Mandt
NeurIPS, 2024

CMMD: Contrastive Multi-Modal Diffusion for Video-Audio Conditional Modeling

Ruihan Yang, Hannes Gamper and Sebastian Braun
ECCV AVGenL Workshop, 2024

Lossy Image Compression with Conditional Diffusion Model

Ruihan Yang and Stephan Mandt
NeurIPS, 2023

SC2 Benchmark: Supervised Compression for Split Computing

Yoshitomo Matsubara, **Ruihan Yang**, Marco Levorato and Stephan Mandt
TMLR (Journal)

Insights from Generative Modeling for Neural Video Compression

Ruihan Yang, Yibo Yang, Joe Marino and Stephan Mandt
IEEE PAMI (Journal)

Diffusion Probabilistic Modeling for Video Generation

Ruihan Yang, Prakhar Srivastava and Stephan Mandt
Entropy (Journal)

Supervised Compression for Resource-Constrained Edge Computing Systems

Yoshitomo Matsubara, **Ruihan Yang**, Marco Levorato and Stephan Mandt
WACV, 2022

Hierarchical Autoregressive Modeling for Neural Video Compression

Ruihan Yang, Yibo Yang, Joe Marino and Stephan Mandt
ICLR, 2021

PIANOTREE VAE: Structured Representation Learning for Polyphonic Music

Ziyu Wang, Yiyi Zhang, Yixiao Zhang, Junyan Jiang, **Ruihan Yang**, Junbo Zhao and Gus Xia
ISMIR, 2020

Deep Music Analogy Via Latent Representation Disentanglement

Ruihan Yang, Dingsu Wang, Ziyu Wang, Tianyao Chen, Junyan Jiang and Gus Xia
ISMIR, 2019

Inspecting and Interacting with Meaningful Music Representations using VAE

Ruihan Yang, Tianyao Chen, Yiyi Zhang and Gus Xia
NIME, 2019

The complex non-collinear magnetic orderings in Ba₂YO₆: A new approach to tuning spin-lattice interactions and controlling magnetic orderings in frustrated complex oxides

Yue-wen Fang, **Ruihan Yang** and Hanghui Chen
Journal of Physics: Condensed Matter (Journal)

A large modulation of electron-phonon coupling and an emergent superconducting dome in doped strong ferroelectrics

Jiaji Ma, **Ruihan Yang**, and Hanghui Chen
Nature Communications (Journal)